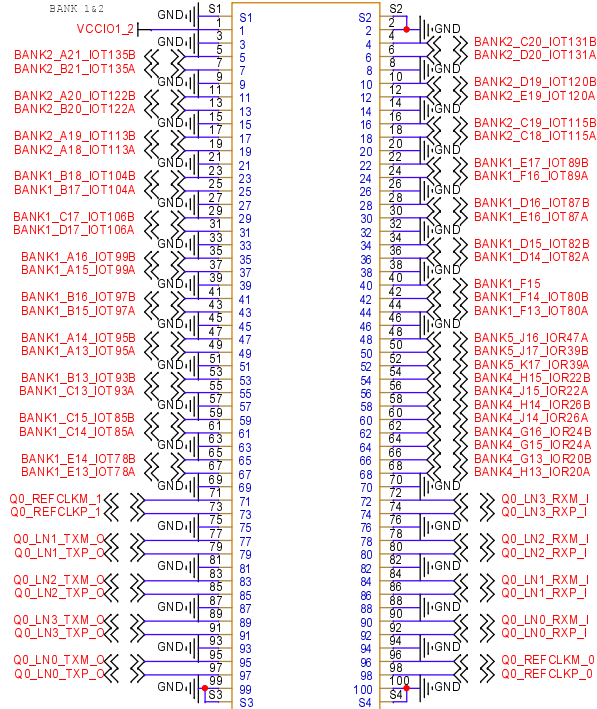
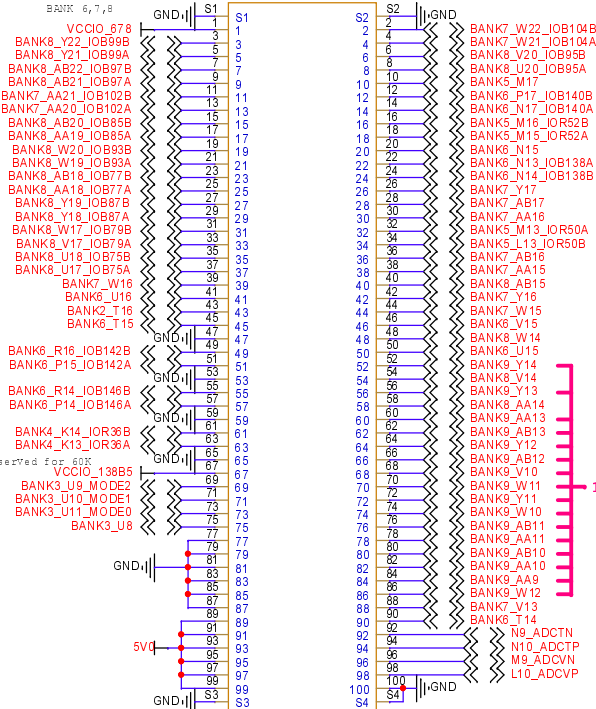


BTB Connector

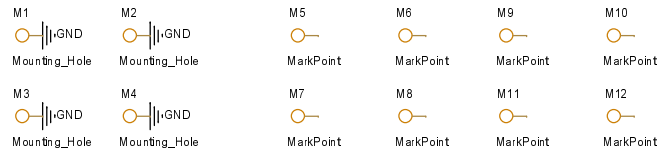
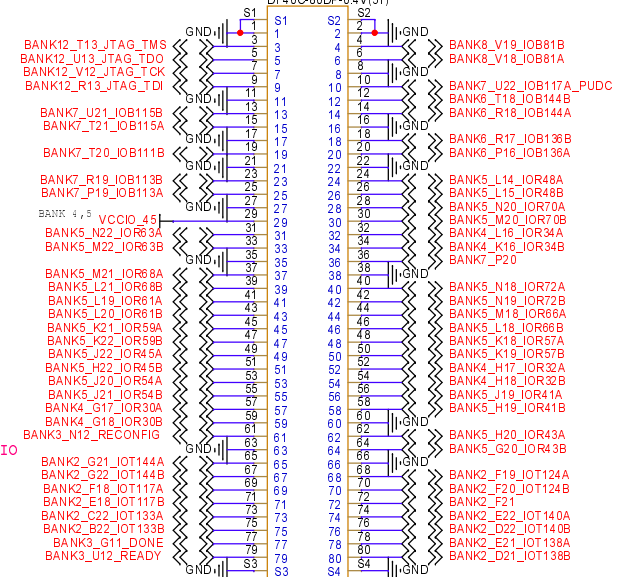
C2399
DF40C-100DP-0.4V(51)



C2400
DF40C-100DP-0.4V(51)

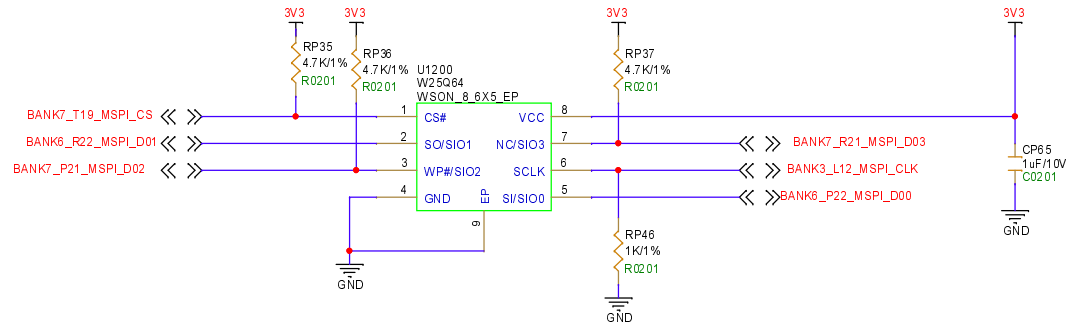


BTB9900
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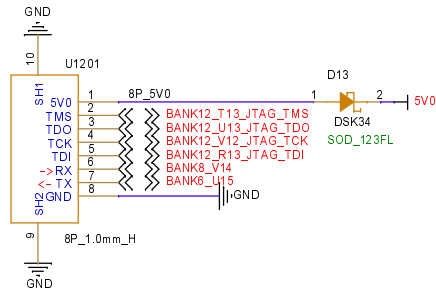


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BTB			
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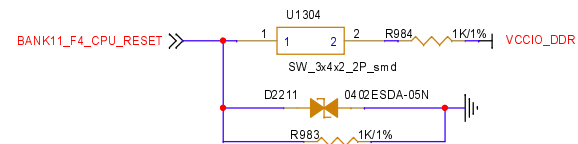
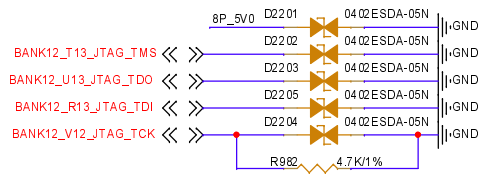
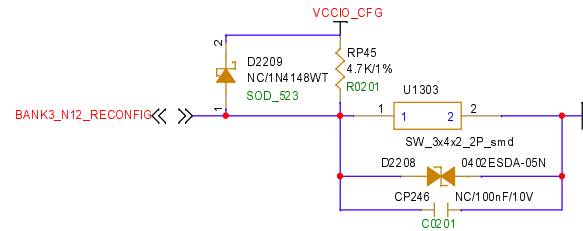
MSPI FLASH



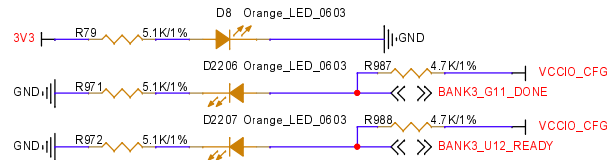
JTAG Connector



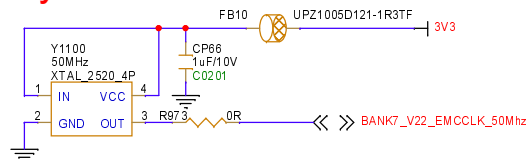
Button



LED



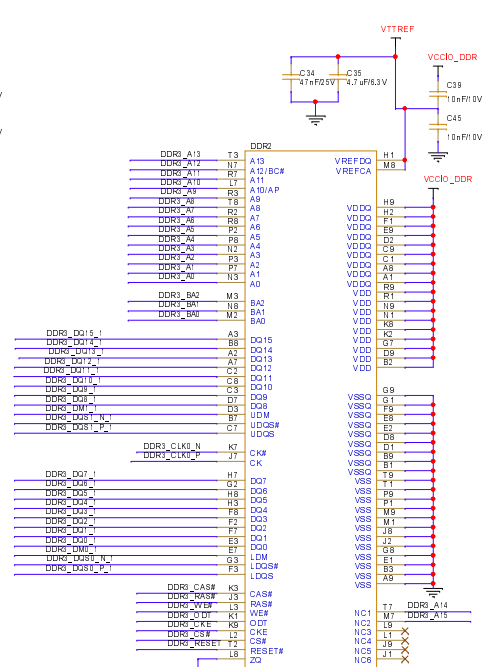
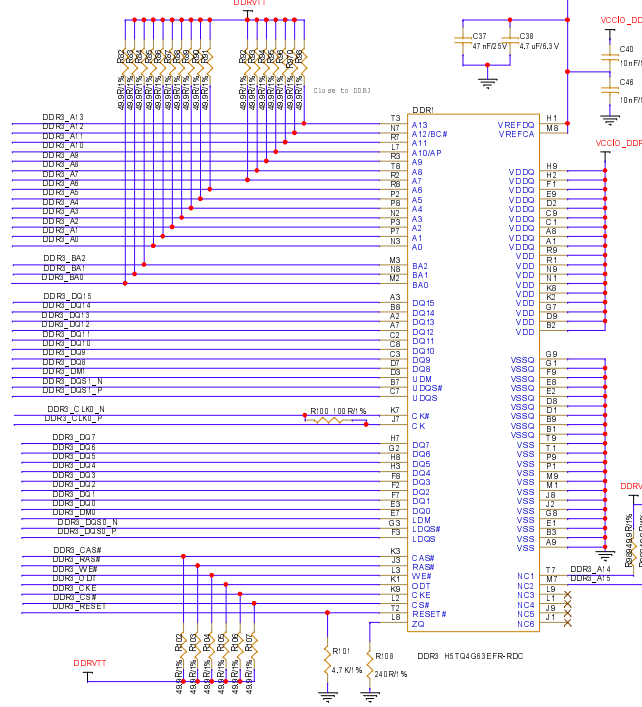
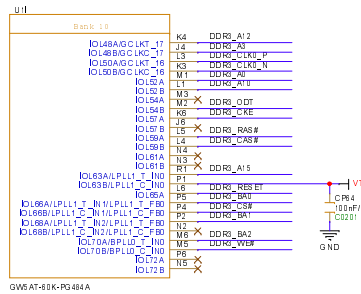
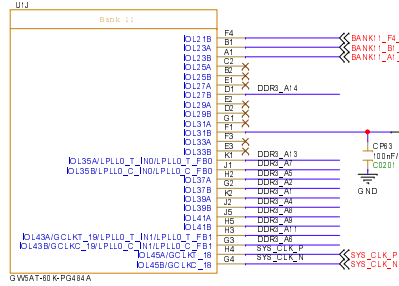
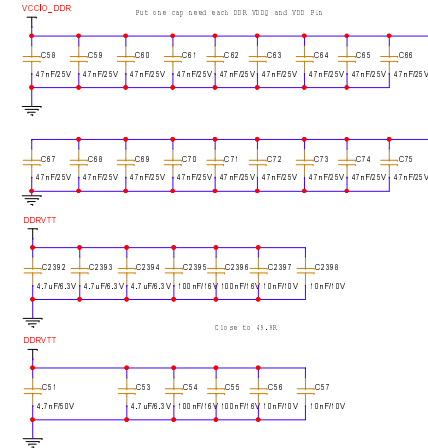
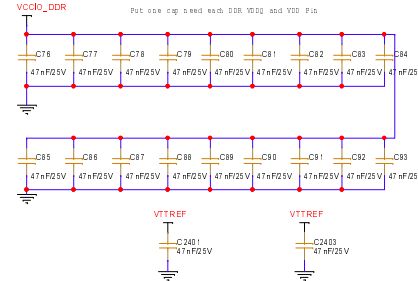
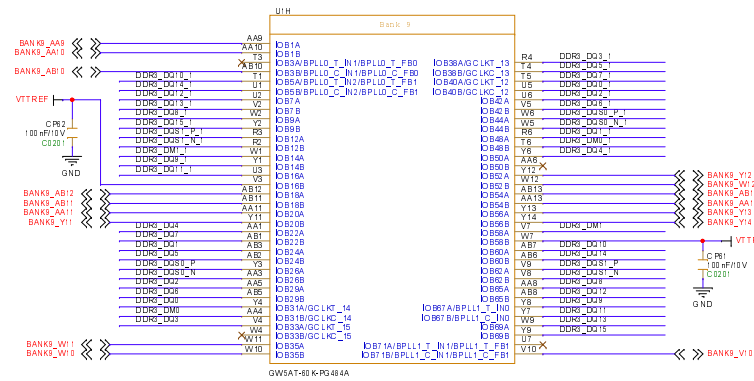
Crystal oscillator



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2 x DDR3

Note:::
BANK 9,10,11 IO Voltage is 1.5V !!!



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BANK 1/2/5

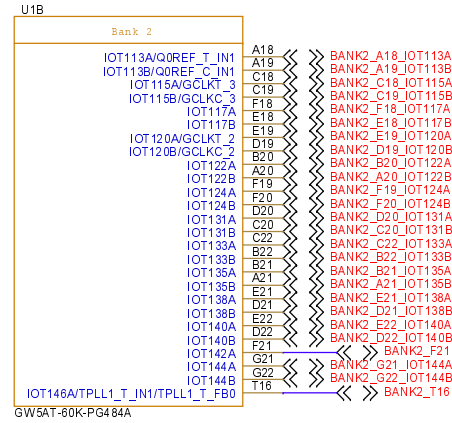
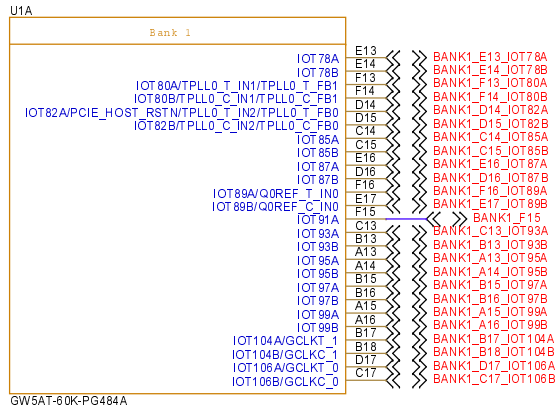
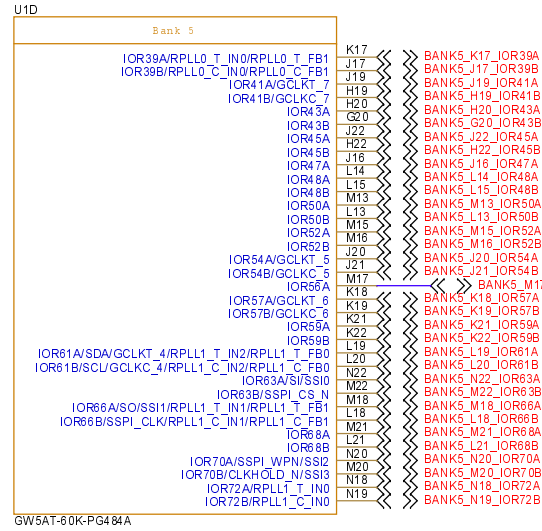
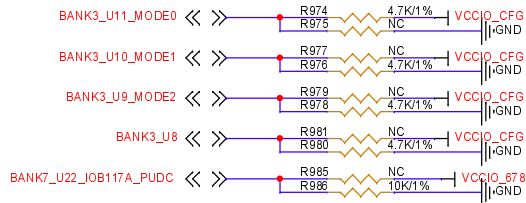


表 4-2 GW5AT-138/GW5AST-138 配置模式选择

配置模式	MODE[2:0] ^[1]	Bus Width	相关说明
JTAG	XXX/101 ^[3]	-	外部 Host 通过 JTAG 接口对 GW5AT & GW5AST 系列 FPGA 产品进行配置
MSPI ^[2]	001	x1,x2,x4	FPGA 作为 Master，通过 SPI 接口从外部 Flash（或其他器件）读取配置数据进行配置
SSPI ^[2]	010	x1,x4	外部 Host 通过 SPI 接口对 GW5AT & GW5AST 系列 FPGA 产品进行配置
Master SERIAL	000	x1	FPGA 作为 Slave 以前，通过 DIN 接口从外部读取配置数据进行配置
Slave SERIAL	111	x1	外部 Host 通过 DIN 接口对 GW5AT & GW5AST 系列 FPGA 产品进行配置
Master CPU	100	x8,x16,x32	FPGA 作为 Slave 以前，通过 DBUS 接口从外部读取配置数据进行配置
Slave CPU	110	x8,x16,x32	外部 Host 通过 DBUS 接口对 GW5AT & GW5AST 系列 FPGA 产品进行配置

注！

- [1]对于一些 MODE 管脚没有全部封装出来的器件，需要查看 PINOUT 手册确认 MODE 脚状态。
- [2] SSPI 和 MSPI 模式的 SPI 接口是互相独立的。
- [3] JTAG 配置模式与 MODE[2:0]输入值无关；当 MODE 设置为 101 时，只有 JTAG 接口生效。



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BANK 3/4/6/7/8/12/GT/ADC

U1C

Bank 4	
IOR20A	H13
IOR20B	G13
IOR22A	H15
IOR22B	H15
IOR24A	G15
IOR24B	G16
IOR26A	J14
IOR26B	H14
IOR30A	G17
IOR30B	G18
IOR32A	H17
IOR32B	H18
IOR34A	L16
IOR34B	K16
IOR36A/RPLL0_T_IN1/RPLL0_T_FB0	K13
IOR36B/RPLL0_C_IN1/RPLL0_C_FB0	K14

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U1E

Bank 6	
IOB128A/D00/MOSI/MI0	P22
IOB128B/D01/DIN/MISO/MI1	R22
IOB130A	V15
IOB130B	U15
IOB132A	T15
IOB132B	T14
IOB134A	U16
IOB134B	N15
IOB136A	P16
IOB136B	R17
IOB138A	N13
IOB138B	N14
IOB140A	N17
IOB140B	P17
IOB142A	P15
IOB142B	R16
IOB144A	T18
IOB144B	P14
IOB146A	R14
IOB146B	

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U1F

Bank 7	
IOB102A/D11	AA20
IOB102B/D12	AA21
IOB104A/D09	W21
IOB104B/D10	W22
IOB106A	AA16
IOB106B	AA15
IOB108A	AB17
IOB108B	AB16
IOB111A/MCS_N	T19
IOB111B/D08	T20
IOB113A/D06	P19
IOB113B/D07	R19
IOB115A/D04	T21
IOB115B/D05	U21
IOB117A/PUDC_B	U22
IOB117B/EMCCLK	V22
IOB120A/D02/MI2	P21
IOB120B/D03/MI3	R21
IOB122A	Y17
IOB122B	Y16
IOB124A	P20
IOB124B	V13
IOB126A	W16
IOB126B	W15

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U1G

Bank 8	
IOB75A	U17
IOB75B	U18
IOB77A/BPLL1_T_IN2/BPLL1_T_FB0	AA18
IOB77B/BPLL1_C_IN2/BPLL1_C_FB0	AB18
IOB79A/CSI_B	V17
IOB79B	W17
IOB81A/GCLKT_11	V18
IOB81B/GCLKC_11	V19
IOB85A/RDWR_B	AA19
IOB85B/CSO_B/DOUT	AB20
IOB87A/GCLKT_10	Y18
IOB87B/GCLKC_10	Y19
IOB89A	AB15
IOB89B	AA14
IOB91A	W14
IOB91B	W19
IOB93A/GCLKT_9	W20
IOB93B/GCLKC_9	U20
IOB95A/GCLKT_8	V20
IOB95B/GCLKC_8	AB21
IOB97A/DT4	AB22
IOB97B/D15	Y21
IOB99A	Y22
IOB99B/D13	

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U1K

Bank 3&12	
IOR11A/TPLL1_T_IN0/TPLL1_T_FB1	U8
IOR11B/MODE2/TPLL1_C_IN0/TPLL1_C_FB1	L12
IOR5A/CCLK	U9
IOR5B/READY	U12
IOR7B/MODE0	U11
IOR7A/MODE1	U10
IOR9A/DONE	G11
IOR9B/RECONFIG_N	N12
IOR1A/TCK	V12
IOR3B/TDO	U13
IOR1B/TDI	R13
IOR3A/TMS	T13

Bank ADC	
ADCTN	N9
ADCTP	M9
ADCVN	L10
ADCVP	

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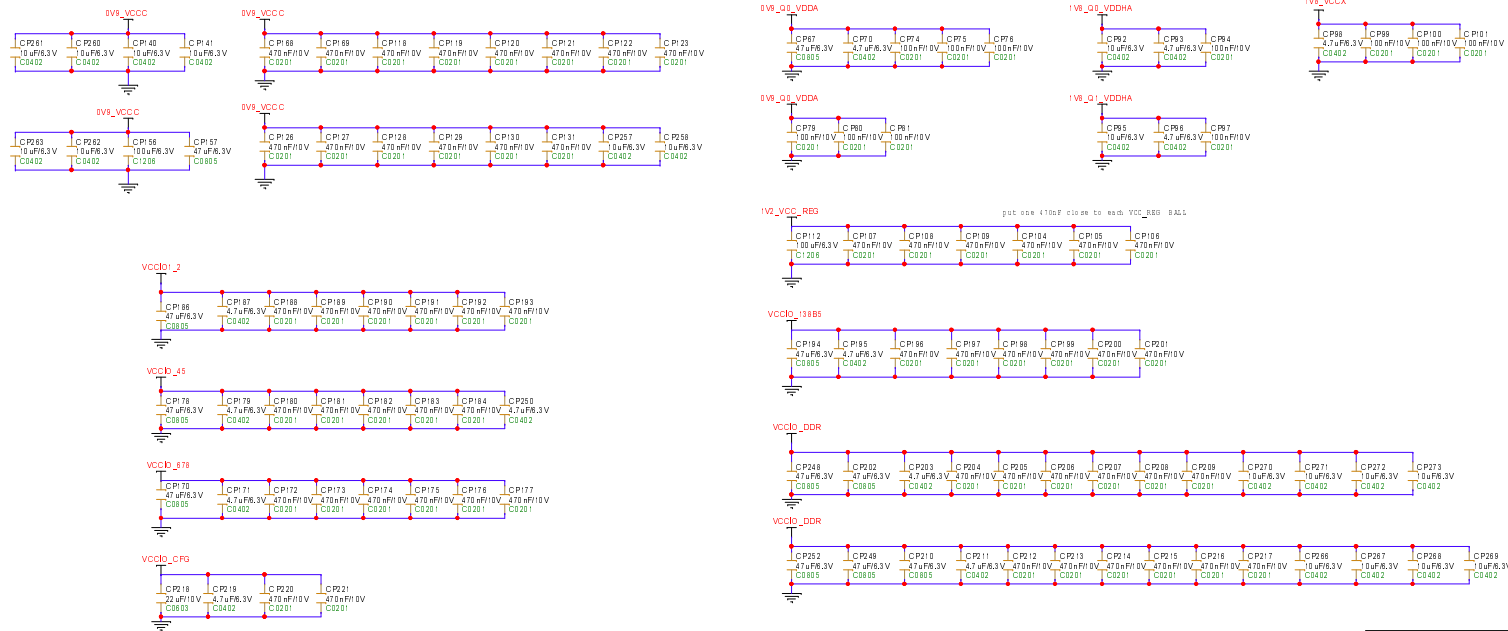
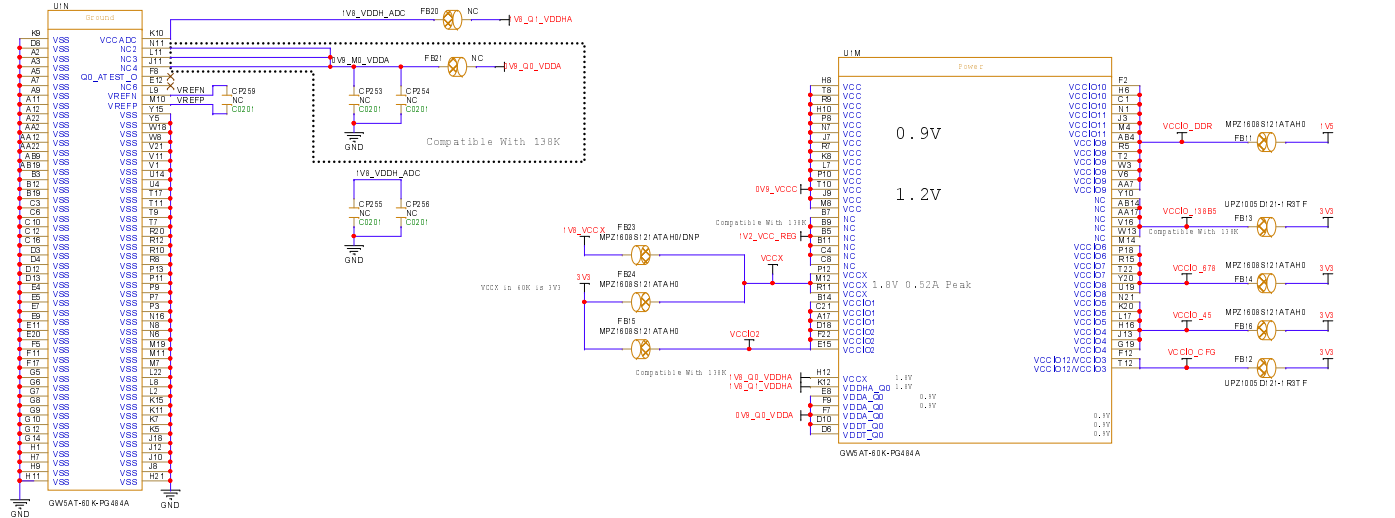
U1L

SerDes Bank Q0	
Q0_LN0_RXM_I	A8
Q0_LN0_RXP_I	B8
Q0_LN0_TXM_O	A4
Q0_LN0_TXP_O	B4
Q0_LN1_RXM_I	C9
Q0_LN1_RXP_I	D9
Q0_LN1_TXM_O	C7
Q0_LN1_TXP_O	D7
Q0_REFCLKM_0	E6
Q0_REFCLKM_1	E10
Q0_REFCLKP_0	F6
Q0_REFCLKP_1	F10

GW5AT-60K-PG484A

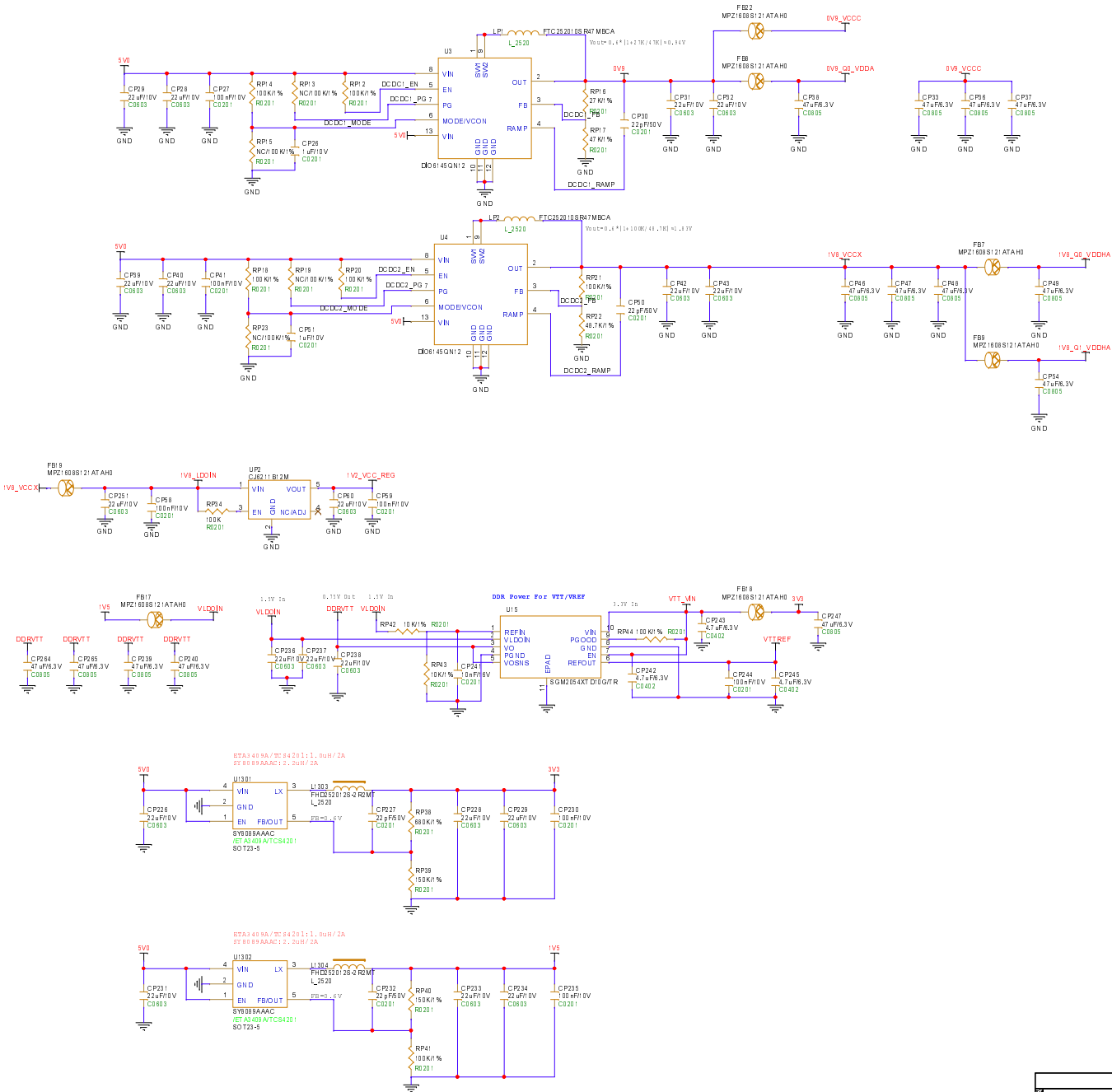
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FPGA POWER



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Board Power



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